Rocuronium

Mechanism of action

- Rocuronium is a neuromuscular blocker. It antagonises (blocks) nicotinic acetylcholine receptors at the neuromuscular junction (motor nerve end plate) of skeletal muscle. This results in the inability of skeletal muscles to contract.

Authorised practice levels

- Only CCPs may administer rocuronium for the purposes of rapid sequence intubation (RSI).
- All ICPs may administer rocuronium post intubation, provided the endotracheal tube position has been confirmed with capnography.

Indications

- ✓ Neuromuscular blockade for RSI.
- ✓ Neuromuscular blockade following endotracheal intubation:
  - » Rocuronium is routinely administered following RSI if suxamethonium is used as the neuromuscular blocker.
  - » Rocuronium is administered if the patient shows clinically significant signs of moving following intubation without RSI.
- ✓ Patient movement during cardiac arrest that is interferring with resuscitation, despite ketamine administration, provided the patient has been intubated and the endotracheal tube position has been confirmed with capnography.

Contraindications

- ✕ Known severe allergy.
- ✕ Endotracheal tube position has not been confirmed by capnography (this does not apply if the indication is for RSI).

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Cautions

- Predicted difficult intubation (this only applies if the indication is for RSI).
- If the indication is for neuromuscular blockade following endotracheal intubation:
  - Chronic muscle weakness. Examples include myasthenia gravis, motor neuron disease and muscular dystrophy. Rocuronium may cause prolonged muscle weakness and should be withheld if possible. If rocuronium is administered the dose should be halved.
  - An adult with a very poor prognosis. Examples include severe comorbidities requiring long-term care, unwitnessed cardiac arrest and cardiac arrest with a first rhythm of asystole. Rocuronium should be withheld if possible because the patient is unlikely to benefit from admission to an intensive care unit and may be extubated in the ED.

Use in pregnancy or when breastfeeding

- Safe and should be administered when indicated.

Dosage

- For RSI: see the 'RSI' guideline.
- For neuromuscular blockade following endotracheal intubation:
  - 100 mg IV for an adult weighing greater than 80 kg.
  - 50 mg IV for an adult weighing less than or equal to 80 kg.
  - See the paediatric drug dose tables for a child.
  - Repeat as required.

Administration

- Administer IV as a bolus.

Common adverse effects

- None.

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Usual onset of effect

- 30-60 seconds. This is predominantly affected by cardiac output and will be prolonged if cardiac output is low.

Usual duration of effect

- 30-60 minutes.

Usual preparation

- Ampoule containing 50 mg in 5 ml.

Pharmacokinetics

- Rocuronium is metabolised in the liver and excreted in urine.
- Significant hepatic or renal impairment will delay clearance and prolong the duration of effect.

Common interactions

- None.

Additional information

- Rocuronium that is not within kits must be refrigerated:
  - Without refrigeration it loses approximately 5-10% of activity per month.
  - It may be safely stored in kits provided it is labelled with the discard date which is eight weeks after removal from the fridge.
  - It may be transported for up to 12 hours between stores/stations in a container that keeps it cool, without affecting the expiry date, provided it is placed into a fridge immediately on arrival.
- Rocuronium is a non-depolarising neuromuscular blocker. This means that the acetylcholine receptor at the neuromuscular junction is not stimulated prior to being blocked and no fasciculations will occur. This is in contrast to suxamethonium which is a depolarising neuromuscular blocker. Suxamethonium binds to the acetylcholine receptor at the neuromuscular junction and stimulates the receptor before blocking it, resulting in brief muscle contractions which are seen as fasciculations.

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- During cardiac arrest the blood flow generated by CPR may be insufficient to deliver adequate levels of rocuronium to skeletal muscle and neuromuscular blockade may not occur. If sustained ROSC is achieved, neuromuscular blockade will occur and the patient will require an adequate level of sedation.
- Adequate sedation and analgesia must always be administered to a patient if rocuronium is administered. See the **'post intubation'** guideline.

More Information

Version history

22 November 2022 1.0.1

Update: Discard date following removal from the fridge.